

ABUBAKER ATTIQUE

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer and Computer Science graduate pursuing an M.Sc. in Artificial Intelligence in Germany, with 2+ years of professional software development experience. Specialized in designing, training, and deploying deep learning systems across computer vision, speech, and multi-modal applications — including models trained from scratch reaching up to 97% accuracy. Strong in Python, PyTorch, TensorFlow, and full-stack development (ReactJS, Flask), with a proven record of shipping production-ready ML applications end to end. Seeking ML/AI Engineer roles where I can build intelligent systems at scale.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, PHP, HTML/CSS

ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, LSTMs, Neural Networks, Computer Vision, MediaPipe, OpenCV, Librosa, Model Training & Evaluation

Data: Pandas, NumPy, Matplotlib, Seaborn, Data Preprocessing, Feature Engineering (MFCC)

Web / Backend: ReactJS, React Native, Flask, REST APIs, PostgreSQL, MongoDB

Tools & DevOps: Git, Docker, Kubernetes, AWS, VS Code, Android Studio, Linux

PROFESSIONAL EXPERIENCE

Developer — R2V

May 2025 – May 2026

- Led frontend development and feature expansion for web and GIS applications, owning delivery from requirements to production deployment.
- Refactored legacy React code into reusable modular components, improving maintainability and reducing rendering bottlenecks.
- Built interactive dashboards, data filters, and GIS map tools with backend integration, optimizing load times and UI stability.
- Acted as technical coordinator, translating stakeholder requirements into responsive, production-ready features and supporting testing and deployment.

React Front-End Developer — R2V

Oct 2024 – Mar 2025

- Developed responsive, cross-device interfaces in ReactJS and implemented reusable components with robust form validation.
- Partnered with designers to improve UI/UX and frontend performance, and resolved UI defects through systematic debugging.

AI Engineer (Internship) — LIMS-COE

Jun 2024 – Sep 2024

- Trained a CNN-based deep learning model for plant-disease classification, aggregating and standardizing datasets from diverse sources to ensure data integrity.
- Integrated the trained model into the organization's production application, enabling real-time diagnostic capabilities.
- Contributed research and insights to technical working papers and presentations for the engineering team.

React Front-End Developer (Internship) — R2V

Jun 2023 – Aug 2023

- Designed and built the frontend of the Balochistan Land Record MIS (BLRMIS) using React, Tailwind, and Leaflet for interactive mapping.
- Collaborated with a cross-functional team of 4 developers to deliver frontend modules 2 weeks ahead of schedule.

Teaching Assistant, Physics — FAST-NUCES

Sep 2022 – Feb 2023

- Mentored and supported 150+ students through weekly office hours, tutorials, and one-on-one sessions.

KEY PROJECTS

AI Interview Coach — Real-Time Multi-Modal Analysis System

2026

Python, PyTorch, MediaPipe, OpenCV, Librosa, Flask, ReactJS | github.com/AbuBakerAttique/AI-Interview-Coach

- Engineered a multi-modal AI pipeline with 4 custom deep learning models for real-time interview analysis via webcam and microphone.

- Trained a Speech CNN+LSTM from scratch on RAVDESS (2,880 audio files) using MFCC features — 93.4% accuracy; a Facial-Expression CNN on FER2013 (28,821 images) — 75.5%; and a Posture NN on MediaPipe landmark data (4,794 samples) — 97.3%.
- Built a score-fusion layer combining all 4 models into a final confidence score with grade and actionable feedback.

AI Gym Rep Counter & Form Analyzer

2026

Python, MediaPipe, NumPy, Three.js, Web Speech API | github.com/AbuBakerAttique/Gym-Project

- Built a real-time computer-vision fitness tracker using MediaPipe Pose to count reps and detect form errors across 6 exercises — no paid APIs.
- Engineered a joint-angle calculation engine with NumPy and a state-machine rep counter using exercise-specific elbow, knee, and hip thresholds.
- Implemented phone-as-camera mode via MediaPipe.js with self-signed HTTPS, plus a Three.js 3D form guide and real-time voice feedback.

Plant Disease Detection System (AI + Web App)

2024

Python, TensorFlow, CNN, Flask, REST API

- Developed a CNN model to classify plant-leaf diseases from images and served it through a backend inference API for real-time predictions.
- Built a responsive full-stack web app with image preprocessing and confidence scoring for non-technical users (farmers).

Mute Motion — AI Sign-Language to Text

2023 – 2024

React Native, Firebase, Computer Vision | Final-Year Project

- Built an AI-driven sign-language recognition system to enable communication between sign-language and non-sign-language users.
- Implemented a React Native frontend with a Firebase backend for real-time interaction.

EDUCATION

M.Sc. in Artificial Intelligence

Apr 2026 – Present

Brandenburg University of Technology Cottbus-Senftenberg • Cottbus, Germany

B.Sc. in Computer Science

Sep 2020 – Jun 2024

FAST-NUCES (National University of Computer & Emerging Sciences) • Pakistan